ANSWER KEY & MARKING SCHEME SET 1

Chapter 2: **Arithmetic Expressions** | Class 7 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 15	Apply BODMAS: brackets, orders, division/multiplication, addition/subtraction.
2	**(a)** 15	Sum first, then double, then subtract.
3	**(a)** -4	Evaluate both products.
4	**(a)** 2277	Distributive property.
5	**(a)** 100	Distributive property.
6	**(c)** 130	Add the decreasing distances.
7	**(d)** 3	Apply BODMAS: brackets, orders, division/multiplication, addition/subtraction.
8	**(d)** 24	Apply BODMAS: brackets, orders, division/multiplication, addition/subtraction.
9	**(a)** Both A and R are true and R is the correct explanation of A.	3 x 4 = 12 and 5 + 12 = 17.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	Both are true, but R does not explain A.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Innermost brackets; Next level (1); Final operations; Value $= 15$ (1).	1+1
12	$= 37 \times (48 + 52)$ (1); $= 37 \times 100 = 3700$ (1).	1+1
13	First $= 150$; Second $= 154$ (1); First is smaller (1).	1+1
14	Innermost brackets; Next level (1); Final operations; Value $= 82$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Full sets $= 24 \div 6 = 4$; Discount $= 4 \times 20 = 80$ (1); Total $= 24 \times 35 + 30 \times 12 - 80 = 1120$ (1); Check with units (₹) (1).	1+1+1
16	First $= 180$ (1); Second $= (12 + 2) \times (15 - 2) = 182$ (1); Difference $= 2$ (1).	1+1+1
17	Outer area $= (20 + 2 \times 2) \times (12 + 2 \times 2) = 384$ (1); Garden area $= 20 \times 12 = 240$ (1); Path area $= 144$ m^2 (1).	1+1+1
	OR: Full sets $= 30 \div 5 = 6$; Discount $= 6 \times 25 = 150$ (1); Total $= 30 \times 40 + 25 \times 15 - 150 = 1425$ (1); Check with units (₹) (1).	1+1+1
18	First $= 252$ (1); Second $= (14 + 3) \times (18 - 3) = 255$ (1); Difference $= 3$ (1).	1+1+1
	OR: Outer area $= (30 + 2 \times 3) \times (18 + 2 \times 3) = 864$ (1); Garden area $= 30 \times 18 = 540$ (1); Path area $= 324$ m^2 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $3 \times 25 + 2 \times 10$ (1). (ii) $75 + 20 = 95$ (1). (iii) $100 - 95 = ₹5$ (1). (iv) Cost $= 5 \times 25 + 2 \times 10 = 125 + 20 = 145$ (1); change $= 200 - 145 = ₹55$ (1).	1+1+1+2
20	Income $= 50 \times 4 \times 12 + 30 \times 3 \times 20$ (1); $= 4200$ (1); Expenses $= 700 + 500 = 1200$ (1); Profit $= 3000$ (1); Expression uses brackets correctly (1).	1+1+1+1+1
	OR: First $= 120$ (1); Second $= 40$ (1); Larger: first (1); Difference 80 (2).	1+1+1+2

Mistakes to avoid:

- Working strictly left to right, e.g. $5 + 3 \times 4 = 32$; multiplication comes first.
- Doing $12 \div 3 \times 2$ as $12 \div 6$; division and multiplication are done left to right.
- Treating subtraction as associative: $20 - (8 - 4) \neq 20 - 8 - 4$.
- Using the distributive property wrongly: $a \times (b + c) \neq a \times b + c$.
- Forgetting to evaluate the bracket first, e.g. taking $2 \times (9 - 4)$ as $18 - 4$.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 2: **Arithmetic Expressions** | Class 7 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** -6	Evaluate both products.
2	**(d)** 4455	Distributive property.
3	**(a)** 100	Distributive property.
4	**(a)** 94	Add the decreasing distances.
5	**(b)** 7	Apply BODMAS: brackets, orders, division/multiplication, addition/subtraction.
6	**(c)** 21	Apply BODMAS: brackets, orders, division/multiplication, addition/subtraction.
7	**(d)** 32	Apply BODMAS: brackets, orders, division/multiplication, addition/subtraction.
8	**(b)** 27	Sum first, then double, then subtract.
9	**(c)** A is true but R is false.	(20 - 8) - 4 = 8, so A is true. 20 - (8 - 4) = 16, so R is false.
10	**(d)** A is false but R is true.	12 / 3 x 2 = 4 x 2 = 8, so A is false. R is true.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$= 64 \times (27 + 73)$ (1); $= 64 \times 100 = 6400$ (1).	1+1
12	First $= 294$; Second $= 300$ (1); First is smaller (1).	1+1
13	Innermost brackets; Next level (1); Final operations; Value $= 32$ (1).	1+1
14	$= 25 \times (66 + 34)$ (1); $= 25 \times 100 = 2500$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Full sets $= 36 \div 6 = 6$; Discount $= 6 \times 30 = 180$ (1); Total $= 36 \times 28 + 40 \times 10 - 180 = 1228$ (1); Check with units (₹) (1).	1+1+1
16	First $= 130$ (1); Second $= (10 + 1) \times (13 - 1) = 132$ (1); Difference $= 2$ (1).	1+1+1
17	Outer area $= (25 + 2 \times 2) \times (15 + 2 \times 2) = 551$ (1); Garden area $= 25 \times 15 = 375$ (1); Path area $= 176$ m^2 (1).	1+1+1
	OR: Full sets $= 20 \div 4 = 5$; Discount $= 5 \times 15 = 75$ (1); Total $= 20 \times 45 + 18 \times 20 - 75 = 1185$ (1); Check with units (₹) (1).	1+1+1
18	First $= 320$ (1); Second $= (16 + 4) \times (20 - 4) = 320$ (1); Difference $= 0$ (1).	1+1+1
	OR: Outer area $= (40 + 2 \times 4) \times (24 + 2 \times 4) = 1536$ (1); Garden area $= 40 \times 24 = 960$ (1); Path area $= 576$ m^2 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $12 \times (18 + 12)$ (1). (ii) $12 \times 30 = 360$ (1). (iii) $12 \times 18 + 12 \times 12 = 216 + 144 = 360$ (1). (iv) $12 \times 98 = 12 \times (100 - 2) = 1200 - 24 = 1176$ (2).	1+1+1+2
20	Income $= 70 \times 3 \times 8 + 35 \times 4 \times 22$ (1); $= 4760$ (1); Expenses $= 650 + 450 = 1100$ (1); Profit $= 3660$ (1); Expression uses brackets correctly (1).	1+1+1+1+1
	OR: First $= 4$ (1); Second $= 16$ (1); Larger: second (1); Difference 12 (2).	1+1+1+2

Mistakes to avoid:

- Working strictly left to right, e.g. $5 + 3 \times 4 = 32$; multiplication comes first.
- Doing $12 \div 3 \times 2$ as $12 \div 6$; division and multiplication are done left to right.
- Treating subtraction as associative: $20 - (8 - 4) \neq 20 - 8 - 4$.
- Using the distributive property wrongly: $a \times (b + c) \neq a \times b + c$.
- Forgetting to evaluate the bracket first, e.g. taking $2 \times (9 - 4)$ as $18 - 4$.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 3

Chapter 2: **Arithmetic Expressions** | Class 7 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 100	Distributive property.
2	**(d)** 165	Add the decreasing distances.
3	**(a)** 4	Apply BODMAS: brackets, orders, division/multiplication, addition/subtraction.
4	**(a)** 2	Apply BODMAS: brackets, orders, division/multiplication, addition/subtraction.
5	**(b)** 6	Apply BODMAS: brackets, orders, division/multiplication, addition/subtraction.
6	**(b)** 27	Sum first, then double, then subtract.
7	**(a)** -8	Evaluate both products.
8	**(a)** 7722	Distributive property.
9	**(a)** Both A and R are true and R is the correct explanation of A.	800 - 8 = 792, using the distributive property.
10	**(a)** Both A and R are true and R is the correct explanation of A.	9 - 4 = 5, 2 x 5 = 10 and 6 + 10 = 16.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	First $= 510$; Second $= 522$ (1); First is smaller (1).	1+1
12	Innermost brackets; Next level (1); Final operations; Value $= 6$ (1).	1+1
13	$= 18 \times (45 + 55)$ (1); $= 18 \times 100 = 1800$ (1).	1+1
14	First $= 216$; Second $= 221$ (1); First is smaller (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Full sets $= 48 \div 8 = 6$; Discount $= 6 \times 40 = 240$ (1); Total $= 48 \times 32 + 36 \times 14 - 240 = 1800$ (1); Check with units (₹) (1).	1+1+1
16	First $= 187$ (1); Second $= (11 + 2) \times (17 - 2) = 195$ (1); Difference $= 8$ (1).	1+1+1
17	Outer area $= (36 + 2 \times 3) \times (20 + 2 \times 3) = 1092$ (1); Garden area $= 36 \times 20 = 720$ (1); Path area $= 372$ m^2 (1).	1+1+1
	OR: Full sets $= 42 \div 7 = 6$; Discount $= 6 \times 35 = 210$ (1); Total $= 42 \times 38 + 24 \times 16 - 210 = 1770$ (1); Check with units (₹) (1).	1+1+1
18	First $= 247$ (1); Second $= (13 + 3) \times (19 - 3) = 256$ (1); Difference $= 9$ (1).	1+1+1
	OR: Outer area $= (45 + 2 \times 5) \times (30 + 2 \times 5) = 2200$ (1); Garden area $= 45 \times 30 = 1350$ (1); Path area $= 850$ m^2 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $8 + 12 - 2 = 18$ (1). (ii) $12 \times 3 - 2 = 34$ (1). (iii) $8 + 4 \times 1 = 12$ (1). (iv) $(8 + 4) \times 3 - 2 = 34$ is the greatest (1); $34 - 18 = 16$ more (1).	1+1+1+2
20	Income $= 55 \times 5 \times 6 + 25 \times 4 \times 28$ (1); $= 4450$ (1); Expenses $= 600 + 350 = 950$ (1); Profit $= 3500$ (1); Expression uses brackets correctly (1).	1+1+1+1+1
	OR: First $= 64$ (1); Second $= 24$ (1); Larger: first (1); Difference 40 (2).	1+1+1+2

Mistakes to avoid:

- Working strictly left to right, e.g. $5 + 3 \times 4 = 32$; multiplication comes first.
- Doing $12 \div 3 \times 2$ as $12 \div 6$; division and multiplication are done left to right.
- Treating subtraction as associative: $20 - (8 - 4) \neq 20 - 8 - 4$.
- Using the distributive property wrongly: $a \times (b + c) \neq a \times b + c$.
- Forgetting to evaluate the bracket first, e.g. taking $2 \times (9 - 4)$ as $18 - 4$.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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